

Stock Assessment Model Report

Samir Johnson

February 2025

1 Introduction

This report details the development and evaluation of machine learning models for classifying the sustainability status of global fish stocks. Accurate stock assessment is crucial for effective fisheries management and the long-term health of marine ecosystems. We explore the use of time series data, including Catch per Unit Effort (CPUE) and various oceanographic parameters, to predict whether a stock is sustainably or unsustainably fished. The report outlines the data sources and aggregation methods, the feature selection process to mitigate multicollinearity, and the architecture and performance of univariate and multivariate MultiRocket Transformer models. While we demonstrate the ability to build a lightweight model for stock classification with limited data, our findings also underscore the inherent limitations that stem from low resolution of training data available.

Project code can be found at this git repository:

<https://github.com/samirjohnson48/ML-Stock-Assessment>.

2 Datasets

2.1 Catch and Effort

2.1.1 Definitions and Sources

The strongest covariate for assessing stock status is Catch per Unit Effort (CPUE), defined as

$$CPUE = \frac{C}{E} \tag{1}$$

where C , E are catch and effort, respectively. We can rewrite the catch C in terms of the fishing effort, fish population density D , and catchability coefficient q as

$$C = qED \tag{2}$$

which gives

$$CPUE = \frac{C}{E} = qD \quad (3)$$

That is, the CPUE contains information on the fish population within a given area and the catchability of the fish population.

We obtain catch data from FishStatj, an application developed by the FAO for the purposes of publishing their statistics on Fisheries and Aquaculture. We utilize the global production workspace to obtain capture production data from 1950 - 2022, with capture reported by location (FAO major fishing area), country responsible for capture, and species.

2.1.2 Aggregation

The first aggregation procedure we perform on the data is calculating CPUE for a specific species within a given FAO major fishing area. Capture data is reported by area and country, however the specific stock which is reported in the capture is not known from the dataset. Even if a certain country is responsible for the capture, it is not guaranteed the stock which is caught corresponds to the stock near that country. Sometimes there are even multiple stocks per country (e.g. Brazil N, Brazil S). Thus, we aggregate the capture data by species and area. We perform the same operation for effort, aggregating the country specific reported fishing effort over FAO major fishing areas. This gives the final aggregate CPUE as

$$CPUE_{t,s,A} = \frac{\sum_{c \in A} C_{t,s,c}}{\sum_{c \in A} E_{t,c}} \quad (4)$$

where t is the year, s is the species, A is the area, and c are the countries within area A .

2.1.3 Effort extrapolation

The fishing effort data collected only reaches until 2015, while the stock assessment data from SOFIA to be used reaches until 2021. Thus, modeling the effort past 2015 is a necessary procedure in order to retain a significant portion of the training data to be used in the model.

We use a simple linear extrapolation model on the previous 20 data points in the effort data. A simple assumption is made that fishing effort does not decrease from 2015 onward, which overrules the linear extrapolation for predicted negative slopes.

2.2 Oceanographic Data

2.2.1 Definitions and Sources

Oceanographic data was collected from Copernicus Marine Data Store via two products: Global Ocean Ensemble Physics Reanalysis (<https://doi.org/10.48670/moi-00024>) and Global Ocean Biogeochemistry Hindcast (<https://doi.org/10.48670/moi-00019>). From these data products, we used specific datasets, listed below in Table 1. Both of these datasets provided oceanographic data on a resolution of 0.25° with 75 vertical levels from 1993 onward. We list the variables used in the model below from both datasets, along with their descriptions. For variables from the Global Ocean Ensemble Physics Reanalysis, we use the reported values which are averages over the data ensemble used. These data sources are GLORYS2V4, C-GLORS, and ORAS5.

CM Data Product	Dataset
Global Ocean Ensemble Physics Reanalysis	cmems_mod_glo_bgc_my_0.25deg_P1M-m
Global Ocean Biogeochemistry Hindcast	cmems_mod_glo_bgc_my_0.25deg_P1M-m

Table 1: Copernicus Marine product from Data Store and corresponding dataset used from product.

Variable (Name in Data)	CM Data Product	Description
<code>thetao_mean</code>	Physics	Mean of Input Analyses Potential Temperatures ($^{\circ}\text{C}$)
<code>so_mean</code>	Physics	Mean of Input Analyses Salinities
<code>chl</code>	Biogeochemistry	Mass concentration of Chlorophyll-a
<code>no3</code>	Biogeochemistry	Molar concentration of nitrate
<code>po4</code>	Biogeochemistry	Molar concentration of phosphate
<code>si</code>	Biogeochemistry	Molar concentration of silicate
<code>nppv</code>	Biogeochemistry	Net primary production of biomass expressed as carbon per unit volume
<code>o2</code>	Biogeochemistry	Molar concentration of dissolved molecular oxygen
<code>fe</code>	Biogeochemistry	Molar concentration of dissolved iron
<code>phyc</code>	Biogeochemistry	Molar concentration of phytoplankton expressed as carbon
<code>ph</code>	Biogeochemistry	Sea water pH reported on total scale
<code>spco2</code>	Biogeochemistry	Surface partial pressure of carbon dioxide

Table 2: Oceanographic variables used in feature analysis for model with source and descriptions. Data Product Physics refers to Global Ocean Ensemble Physics Reanalysis while Biogeochemistry refers to Global Ocean Biogeochemistry Hindcast

2.2.2 Aggregation

In order to predict stock assessment using the oceanographic covariates, we aggregated the oceanographic data by FAO major fishing area. We define approximate regions by a set of coordinates in longitude, latitude space to match the regions defined by the FAO areas, shown in the figure below.

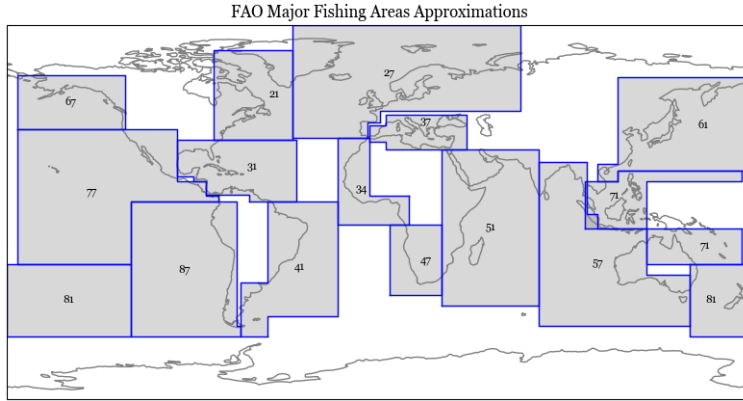


Figure 1: Regions used for oceanographic data aggregation to approximate the FAO major fishing areas.

2.3 Training Data: SOFIA 2024

2.3.1 Definitions and Sources

The target data is taken from the State of World Fisheries and Aquaculture (SOFIA) 2024 dataset, containing stock assessments from 2009 - 2021 for 249 unique species across the various FAO major fishing areas. The stocks are classified in three categories: Underfished (U), Fully Fished (F), and Overfished (O).

The stock assessments are listed within the data in various ways:

- Species with single stock within a given area will have a single classification, e.g. F
- Species with multiple stocks within a given area will have multiple classifications (separated by ", " or ";"), e.g. U, F, U
- Species with single stock and multiple assessments will have multiple classifications (separated by "- ", "/" or "/"), with most accurate classification listed first, e.g. O-F-O

We simplified the classification into binary classification for sustainably fished (either underfished or fully fished) and unsustainably fished (overfished). Since the data are presented in such a way that the distinct stocks are unknown, we aggregated the data across the stocks to create a binary classification for each species within each area. The algorithm for classifying the stocks is listed in the Appendix.

3 Feature Selection

In order to avoid multicollinearity in our model as much as possible, we must determine which environmental covariates are too correlated in order to use as input. We visualize the Pearson correlation matrix of the environmental variables below:

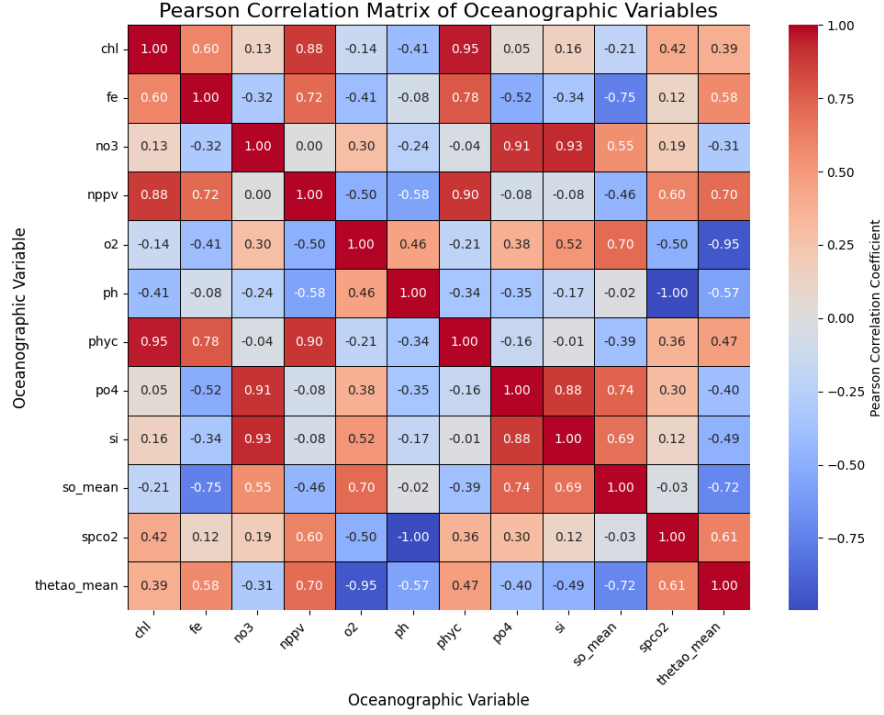


Figure 2: Pearson correlation matrix for all oceanographic variables to be potentially used in the multivariate model. Highly correlated variables are highlighted in red.

Using the heuristic of Pearson correlation coefficient $> 0.7 \rightarrow$ highly correlated, we use the following set of variables as input to our multivariate model:

- thetao_mean
- nppv
- spco2
- no3

4 Models

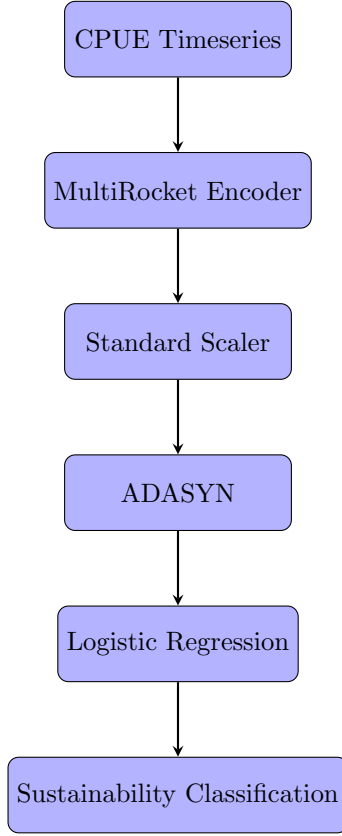
4.1 Encoder: MultiRocket Transformer

The MultiRocket (Multiple Random Convolutional Kernels for Time Series Classification) encoder is a powerful and efficient technique for feature extraction from time series data. It operates by convolving a large number of randomly generated convolutional kernels of varying lengths and weights with the input time series. Each convolution produces a set of feature maps, which are then globally max-pooled to generate a single feature value for each kernel. This process effectively captures a diverse range of temporal patterns and dependencies present in the time series. The resulting set of features, derived from hundreds or thousands of these random convolutions, forms a robust representation of the original time series, suitable for input to downstream machine learning classifiers to build a complete time series classification pipeline. MultiRocket is particularly advantageous due to its computational efficiency, making it applicable to large datasets, and its ability to achieve state-of-the-art performance in time series classification tasks, often outperforming more complex deep learning approaches.

The MultiRocket encoder uses four pooling operators after applying a convolution on the initial time series data for a given kernel: *Proportion of Positive Values* (PPV), *Mean of Positive Values* (MPV), *Mean of Indices of Positive Values* (MIPV), and *Longest Stretch of Positive Values* (LSPV). These pooling operators measure the similarity of a given time series with respect to a given kernel in four distinct ways, capturing the global similarity (PPV, MPV) as well as temporal locations of similarities (MIPV, LSPV). The latter is especially important in time series classification for stock assessment, as temporal locations of trends in data are guaranteed to play a key factor in the health of stocks (e.g. sharp decline in CPUE in 2010's will most likely indicate an overfished stock, while a sustainably fished stock may have a similar decline in the 1960's). This makes MultiRocket a robust and suitable encoder for this classification task.

4.1.1 Univariate MultiRocket Encoder on CPUE

We train a univariate MultiRocket encoder on CPUE data to predict stock assessments. We do this using the following pipeline:



The pipeline uses the MultiRocket encoder as discussed before, which has three main parameters to be tuned for the performance of the model: the number of kernels (nk), the maximum dilation per kernel ($mdpk$), and the number of features per kernel ($nfpk$). Increasing the values of these parameters tends to increase the performance of the model, but at the cost of runtime and available RAM. The pipeline then feeds its computed feature vector into a standard scaler which scales the data down to mean zero and standard deviation of one. Because of the class imbalance inherent in the training data, with approximately 75 percent of the training data classified as sustainably fished and 25 percent classified as unsustainably fished, we utilize the Adaptive Synthetic Sampling Approach (ADASYN) as a unit in the pipeline. This adaptively generates more synthetic data for minority class (unsustainably fished) examples that are harder to learn, meaning those that are closer to the decision boundary or in regions of lower density. Finally, we used a logistic regression classifier, using the liblinear solver due to the smaller training set size (2,500 samples).

We perform a grid evaluation of the univariate model over the available MultiRocket parameters: $nk \in [1000, 5000, 10000]$, $mdpk \in [16, 32]$, and $nfpk \in [4, 8]$. We use a fixed set of parameters for the logistic regression classifying head, setting $C = 0.001$, $max_{iter} = 100$, l2 penalty, *liblinear* solver, and balanced class

weights.

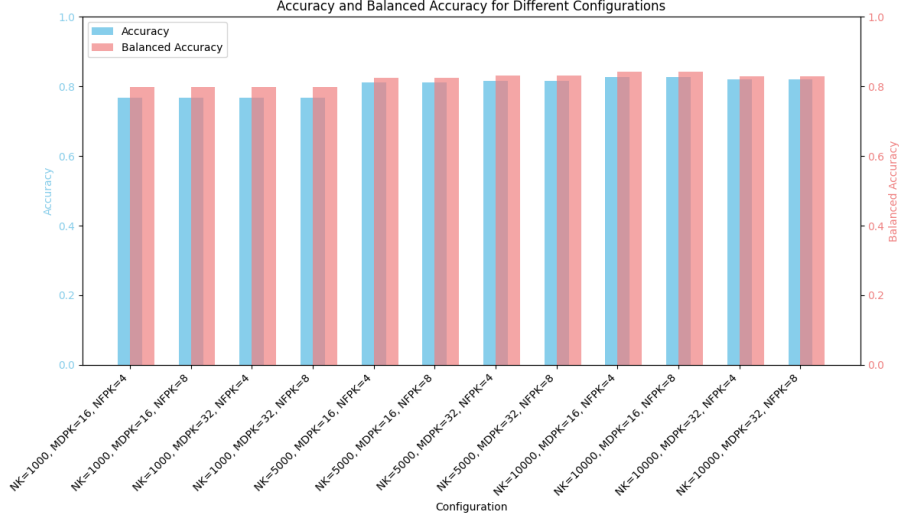
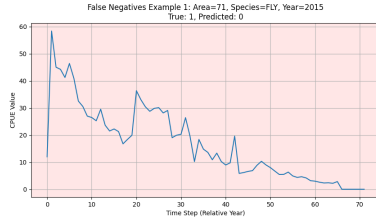


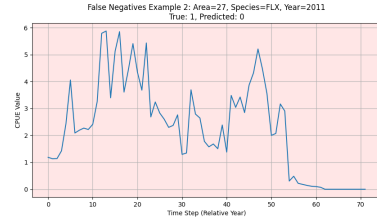
Figure 3: Accuracy and balanced accuracy scores for different configurations of the MultiRocket encoder in the univariate model. Significant improvement can be seen between the models using 1,000 kernels vs. 5,000 kernels (80% vs. 83%, resp.) while marginal improvement is seen between models with 5,000 kernels vs. 10,000 kernels (83% vs. 84%, resp.)

$nk = 10000, mdpk = 16, nfpk = 4$ performed the best with a balanced accuracy score of 0.8417 and a total run time of 59.34 seconds.

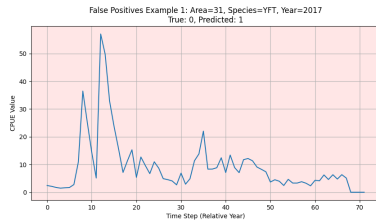
Below are examples of true positives, false positives, true negatives, and false negatives for the classifications of the best performing model. Note that 1 corresponds to sustainably fished and 0 corresponds to unsustainably fished.



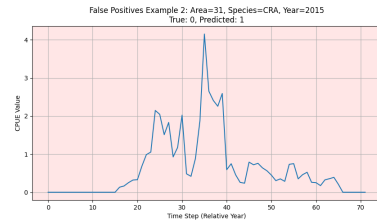
(a) False Negative Example 1



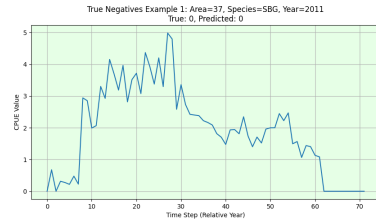
(b) False Negative Example 2



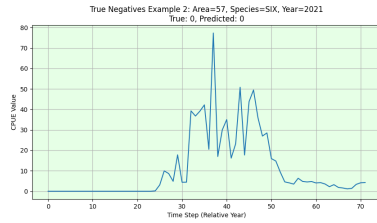
(c) False Positive Example 1



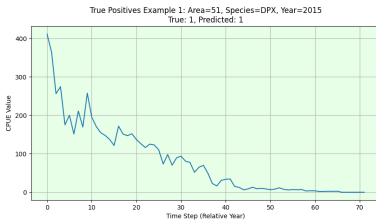
(d) False Positive Example 2



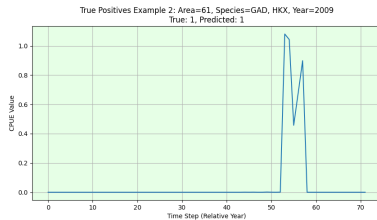
(e) True Negative Example 1



(f) True Negative Example 2



(g) True Positive Example 1



(h) True Positive Example 2

Figure 4: Examples illustrating different classification outcomes for the univariate CPUE model.

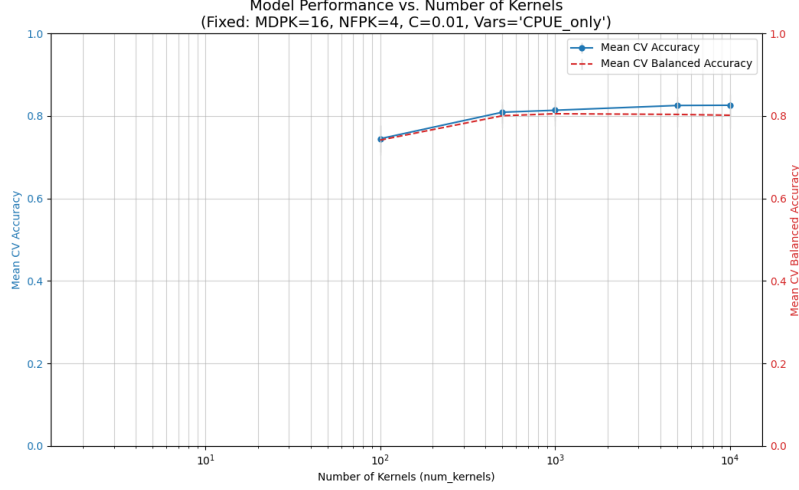
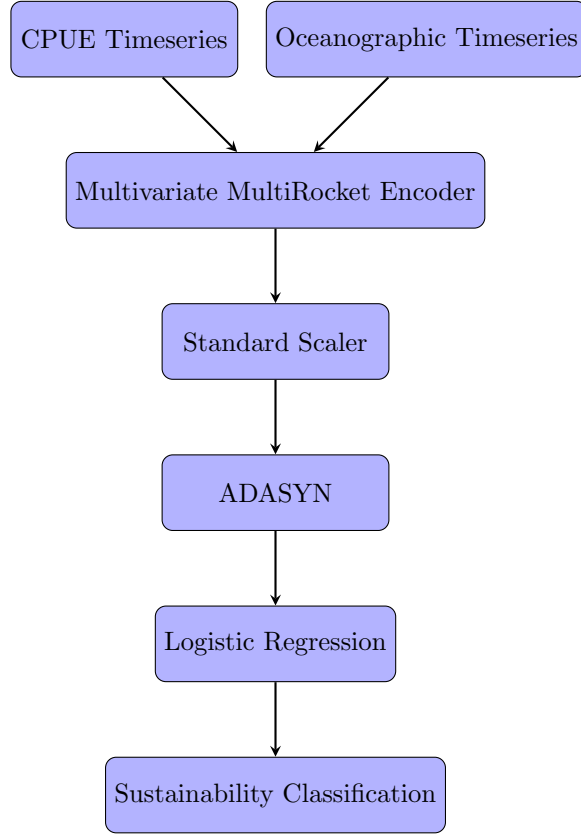


Figure 5: Model performance as a function of number of kernels.

We can see the effect specifically of increasing the number of kernels on the accuracy and balanced accuracy of the univariate model. We use a 10-fold cross validation scheme to decrease the bias of individual fits when evaluating the accuracy and balanced accuracy. We see the accuracy increasing with each increased order of magnitude number of kernels, while the balanced accuracy saturates around 1000 kernels, indicating overfitting beyond this number of kernels with the fixed values of $mdpk = 16$, $nfpk = 4$, $C = 0.01$.

4.1.2 Multivariate MultiRocket Encoder on CPUE and Oceanographic Time Series Data

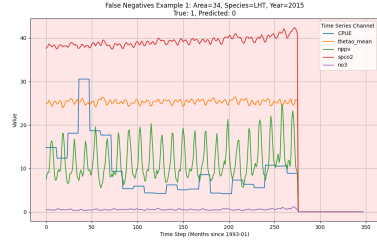
We train a multivariate MultiRocket encoder on CPUE and oceanographic time series data to predict stock sustainability. We do this using the following pipeline:



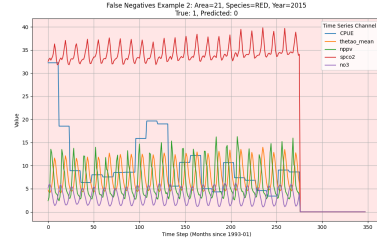
As discussed in the feature selection section, we use the following oceanographic variables: `thetao_mean`, `nppv`, `spco2`, `no3`. These time series variables are only available from 1993, and are reported on a monthly basis from the CM dataset. Thus, we adjust the corresponding CPUE time series to match the oceanographic time series data; we report the CPUE time series data on a monthly basis and begin the series from 1993. To feed the same length time series for each training sample regardless of assessment year, we extend each time series data through 2021 and set the time series to 0 for years after the assessment year. That is, for a stock assessment performed in year y , the time series data for variable v is of the form

$$v_{1993-01}, v_{1993-02}, \dots, v_{y-12}, 0, \dots, 0$$

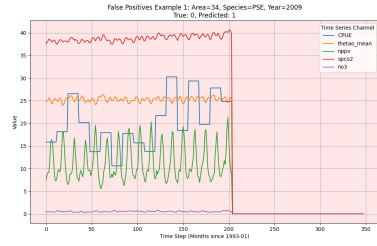
Except for the change from univariate to multivariate MultiRocket encoder, the rest of the pipeline remains the same. Once again, we perform a grid evaluation of the model over the available MultiRocket parameters: $nk \in [1000, 5000, 10000]$, $mdpk \in [8, 16, 32]$, and $nfpk = 4$ (change in $nfpk$ did not change the performance of the model, as seen in the univariate model). We use a fixed set of parameters for the logistic regression classifying head, setting $C = 0.001$, $max_iter = 100$, l2 penalty, *liblinear* solver, and balanced class weights.



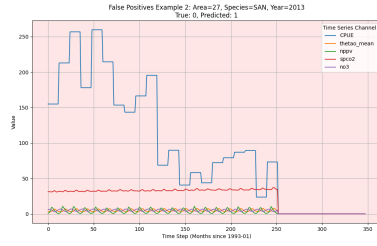
(a) False Negative Example 1



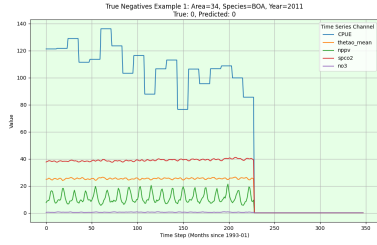
(b) False Negative Example 2



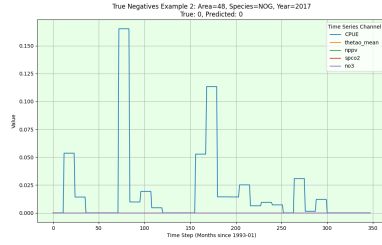
(c) False Positive Example 1



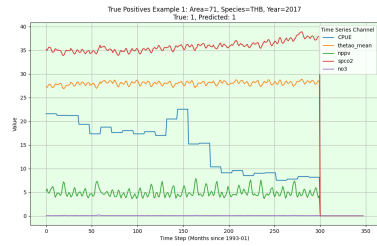
(d) False Positive Example 2



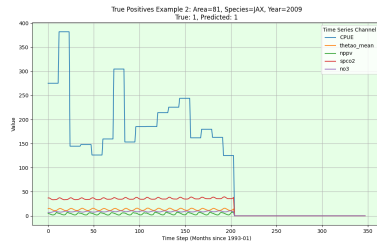
(e) True Negative Example 1



(f) True Negative Example 2



(g) True Positive Example 1



(h) True Positive Example 2

Figure 7: Examples illustrating different classification outcomes for the multi-variate model.

5 Concluding Remarks

Although machine learning techniques are providing more robust and accurate prediction models across many disciplines nowadays, in regards to the current data available in stock assessments, further improvement is needed before implementing these models in this specific discipline. There are three main improvements which need to be made in further collection of stock assessment data which would render such models a viable option moving forward: improved locality of stocks, extension of effort data through the most up-to-date stock assessments, and oceanographic data extending back to at least 1970.

In order to make good use of oceanographic data as inputs in our multi-model prediction model, stocks need to be further localized beyond just the FAO major fishing area. These fishing areas cover large regions of the oceans and seas, and the assumption that data aggregated over these regions can be applied to any stock record in that fishing area does not hold up. By localizing stocks, oceanographic data can be more specific and better reflect the environmental conditions which the stock experiences over time. Development of a universal maritime geo-encoder would facilitate this process, allowing one to convert the standardized name of the maritime location to a set of vertices in longitude, latitude space which define the region. Thus, one could develop a simple data pipeline which takes the name of the location of the stock, converts it to these coordinates in longitude, latitude space, and extracts the necessary environmental data in this region using the Copernicus Marine API or other viable methods.

Localization of stocks would also allow for the use of individual stock assessments, instead of aggregating stock assessments over species for a given FAO area. This would increase the training size and again provide more granular information of stock assessment as a function of CPUE and environmental data.

Effort data through 2021 would help in the accuracy of CPUE as input into the model. Linear extension of effort is sketchy at best, as the fishing effort for each country is quite variable from year to year in different regions. Accurate recent effort data, which in turn create accurate CPUE data for recent years, allows the model to accurately capture the effect of changes in CPUE in recent years on stock assessments. Although its effect on stock assessments results from its trend over many years, and not just its recent trend, it is a viable assumption that its trend in recent years would have the most significant impact on recent stock assessments, especially in cases of drastic changes of CPUE. Thus, accurate CPUE data in recent years is necessary for an accurate model.

Finally, oceanographic data extending back to the 1970's would aid in increasing the accuracy of the multivariate model. With data extending back only to 1993, the CPUE time series data must be truncated as well in order to match the other modes of input. There may be trends crucial in understanding current stock sustainability which extend back before 1993, and these were lost with the truncation of data in the multi-modal model.

6 Appendix

6.1 Target Value Algorithm

```
FUNCTION create_target_values(X)

    X ← X.strip() // Remove leading/trailing whitespace

    IF length(X) = 1 THEN
        IF X is not in ["O", "F", "U"] THEN
            RETURN -1
        ENDIF
        RETURN 1 - (X == "O") // 1 if X is "F" or "U", 0 if X is "O"
    ENDIF

    IF "-" is in X OR "/" is in X THEN
        IF "-" is in X THEN
            sep ← "-"
        ELSE
            sep ← "/"
        ENDIF

        s ← X.split(sep) // Split X by the separator
        FOR each st in s DO
            st ← st.strip()
        ENDFOR

        FOR each st in s DO
            IF st is in ["O", "F", "U"] THEN
                RETURN 1 - (st == "O") // 1 if st is "F" or "U", 0 if st is "O"
            ENDIF
        ENDFOR
        RETURN -1
    ELSIF "," is in X OR ";" is in X THEN
        IF "," is in X THEN
            sep ← ","
        ELSE
            sep ← ";"
        ENDIF

        s ← X.split(sep) // Split X by the separator
        FOR each st in s DO
            st ← st.strip()
        ENDFOR

        IF ("O" is in s AND "F" is in s) OR ("O" is in s AND "U" is in s) THEN
```

```

        RETURN -1
    ELSIF "0" is in s THEN
        RETURN 0
    ELSIF "F" is in s OR "U" is in s THEN
        RETURN 1
    ENDIF
    RETURN -1
ENDIF

RETURN -1

ENDFUNCTION

```

6.2 Effort Extrapolation Algorithm

```

FUNCTION linear_extension(effort):
    idx = MIN(LENGTH(effort), 20)
    data = effort[-(idx):] // Last idx elements of effort

    slope, intercept = np.polyfit(RANGE(idx), data, 1)

    IF slope > 0 THEN
        extension = []
        FOR x IN RANGE(idx, idx + 6):
            value = MAX(slope * x + intercept, 0)
            APPEND value TO extension
        ELSE
            extension = [effort[-1]] * 6 // Repeat last element 6 times

    RETURN CONCATENATE(effort, extension)

```

6.3 Coordinates for FAO Major Fishing Areas Approximations

Area	Coordinates
21	((-80, -42), (35, 78))
27	((-42, -6), (36, 90)), ((-6, 0), (43, 90)), ((0, 68), (49, 90))
31	((-98, -40), (17, 35)), ((-90, -40), (15, 17)), ((-84, -40), (8.4, 15)), ((-63, -40), (5, 8.4))
34	((-20, -5), (-6, 36)), ((-5, 14), (-6, 8))
37	((-5, 3), (34, 42)), ((3, 42), (30, 47))
41	((-67, -54), (-60, -34)), ((-54, -20), (-50, 5))
47	((5, 30), (-40, -6))
51	((30, 77), (-45, 30))
57	((77, 100), (-55, 24)), ((100, 105), (-55, -1)), ((105, 129), (-55, -8)), ((129, 150), (-55, -30))
61	((105, 115), (15, 23)), ((115, 180), (20, 65))
67	((-175, -123), (40, 66))
71	((99, 129), (-8, 15)), ((115, 175), (15, 20)), ((129, 175), (-25, -8))
77	((-175, -120), (-25, 40)), ((-120, -98), (5, 40)), ((-98, -84), (5, 15)), ((-84, -78), (5, 8))
81	((150, 180), (-60, -25)), ((-180, -120), (-60, -25))
87	((-120, -67), (-60, -55)), ((-120, -69), (-55, 5))

Table 3: FAO major fishing area and coordinates for approximate regions. The coordinates are structured as follows: $((longitude_{min}, longitude_{max}), (latitude_{min}, latitude_{max}))$